Chapter 10: **Ramsey Numbers**

10.1 Ordinary Ramsey Numbers

Define $R(m, n)$ as the minimum cardinality N such that every coloring of the edges of the complete graph K_N with red and blue yields either a red K_m or a blue K_n. A trivial example is that $R(2, m) = m$.

EXAMPLE. $R(3, 3) = 6$. *Consider any vertex v in a coloring of K_6. It has at least three edges of same color, say red, going to set S. Then either the set S is all blue, and we're done, or S has a red edge, which extends to a red clique with v. The lower bound is shown by having the red edges be C_5.*

EXAMPLE. $R(3, 4) = 9$. *The lower bound is shown by the circulant $C_8[1, 4]$, which has clique number 2 and independence number 3. To show the upper bound, note that one cannot construct a 3-regular graph on 9 vertices; so there must be either a vertex u with 4 red neighbors or a vertex v with 6 blue neighbors. In the former, we get either a red edge or a blue K_4 in u's red neighborhood. In the latter, we get a monochromatic triangle in v's blue neighborhood.*

The above example proves the idea for the general upper bound:

RECURRENCE. $R(m, n) \leq R(m - 1, n) + R(m, n - 1)$.

(Can be shown to be strict if both terms on the right-hand-side are even.)

Corollary: $R(m, n) \leq \binom{m+n-2}{m-1}$. We show later that $R(m, m) > 2^{m/2}$.

Some more small numbers: $R(3, 5) = 14$ and the extremal graph is the circulant $C_{13}[1, 5]$. $R(4, 4) = 18$ and the extremal graph is the Payley graph on 17 vertices.

10.2 Graph Ramsey Numbers

One can generalize from cliques to general graphs: For graphs G and H, define $R(G, H)$ as the minimum cardinality N such that every coloring of the edges of

K_N with red and blue yields either a red G or a blue H. Note that the copies of G and H do not have to be induced; thus $R(G,H) \le R(m,n)$ where m and n are the orders of G and H.

Here are some elegant results.

Theorem. *For any tree T on m vertices, it holds that $R(T, K_n) = (m-1)(n-1)+1$.*

Proof. Lower bound: Color $K_{(m-1)(n-1)}$ so that the red edges form $(n-1)$ disjoint copies of K_{m-1}.

Upper bound: induction. The base case $m = 2$ or $n = 2$ is easily checked. Now consider an end-vertex u in T with neighbor v. Let $T' = T - u$. By the induction hypothesis we can find either a red T' or a blue K_n. If the latter happens we are done; so assume the former. Let w be the vertex corresponding to v. If w has a red neighbor outside the copy of T', then we are done. So we may assume w has blue-degree at least $(m-1)(n-2)+1$. Now we induct in w's blue neighborhood to get either a red T, or a blue K_{n-1} to which w is joined. ◻

Theorem. *$R(mK_2, mK_2) = 3m - 1$.*

Proof. Lower bound: partition the vertex set of K_{3m-2} into a red clique of size $m-1$ and a blue clique of size $2m-1$ and color red all the edges between the cliques.

Upper bound: Use induction. The base case is $m = 1$ which it trivial. If the coloring of K_{3m-1} is monochromatic, then we are done. So we may assume there exists red and blue edges r and b overlapping in some vertex. Delete the three vertices of r and b, and apply the inductive hypothesis. ◻

10.3 Bipartite Ramsey Numbers

One can further change the host graph. Define the bipartite Ramsey number $B(m,n)$ as the minimum cardinality N such that every coloring of the edges of $K_{N,N}$ with red and blue yields either a red $K_{m,m}$ or a blue $K_{n,n}$. (There are other versions of bipartite Ramsey numbers.)

EXAMPLE. *$B(2,2) = 5$. Lower bound: One can decompose $K_{4,4}$ into two 8-cycles.*

Upper bound: If one colors $K_{5,5}$ there are at least 13 edges of the same color. But it is impossible to have 13 edges in $K_{5,5}$ without a 4-cycle, as we show next.

Suppose one has a subgraph of $K_{5,5}$ with 13 edges but no 4-cycle. Say $K_{5,5}$ has bipartition (X, Y). Let A be the number of P_3's with centers in X. Then $A \leq \binom{5}{2} = 10$, since there can be at most one P_3 for every pair of vertices in Y. On the other hand, $A \geq \sum_{i=1}^{5} \binom{d_i}{2}$, where $d_1, \ldots, d_5$ are the degrees in X. Since these degrees sum to the number of edges, and the binomial coefficient as a function has positive second derivative, the best one can do is degrees $3, 3, 3, 2, 2$, but in any event $A \geq 11$, a contradiction.

The maximum number of edges in a bipartite graph without a specified subgraph is called a ***Zarankiewicz*** number. These turn out to be helpful in bipartite Ramsey theory. Note that the normal equivalents, Turán numbers, are irrelevant to ordinary Ramsey numbers, since the complement of extremal graphs have huge clique number.

The next theorem is about 3-coloring the edges of a complete bipartite graph to avoid a monochromatic 4-cycle. It was proved by Exoo (and later re-proved by GHO).

THEOREM. $B(2,2,2) = 11$.

PROOF. It can be shown that $\lceil 121/3 \rceil = 41$ edges guarantee a 4-cycle in $K_{11,11}$. However, there is a unique subgraph of $K_{10,10}$ that has 34 edges and no 4-cycle. And one can find a suitable coloring of K_{10} (for example using a computer). ⌑